## Supplementary Top-10 Validation Audit

We validated each Top-10 area candidate against documented real-world events using KML and local map context, and cross-checked the event date and location against public news and social-media sources. These labels are external evidence, not labels inferred from detector scores. We report candidate-level precision at several cutoffs; no exhaustive event catalog exists for dataset-wide recall. Repeated rows are grouped conservatively below.

| Evidence | k | Positive | Reported result | Interpretation |
|---|---|---|---|---|
| Area | 1 | 1 | P@1 = 1.00 | Real-world event matching |
| Area | 5 | 5 | P@5 = 1.00 | Real-world event matching |
| Area | 10 | 10 | P@10 = 1.00 | 8 H3-date groups; 3 categories |
| User | 10 | 10 | Map consistency = 10/10 | 4.4--101.5 km shifts; no external cause label |

| Detector ranks | Date/block | Matched real-world event category | Candidates |
|---|---|---|---|
| 1, 5, 7--9 | 28 Jan; Blocks 4--5 | Military demobilization and return travel | 5 |
| 2--4, 6 | 5--6 Feb; Blocks 2--3 | Ancestor-grave gathering, Phu Vinh, Hue | 4 |
| 10 | 28 Jan; Block 5 | Yen Phong interchange congestion | 1 |

These results strengthen top-ranked candidate validation but remain a small, rank-selected sample. They do not estimate recall, population-wide precision, or ten statistically independent events. Figure 2 presents representative area and user maps; Tables VI--VII report all ten ranked area and user candidates.

## Supplementary Shared-Baseline versus LODO Benchmark

Our implemented pipeline materializes one all-sample baseline per comparison context and reuses it. We obtained the one-time baseline-build measurement below from the audited ClickHouse INSERT query. Scoring values are medians of five alternating runs on 26,059,300 corrected-H3 user-blocks. LODO reconstructs a date-excluded reference for every candidate.

| Stage | Frequency | Time | Rows read | Peak memory |
|---|---|---|---|---|
| Shared baseline build | Once | 2.766 s | 26,059,300 | 10.01 GiB |
| Shared-baseline scoring | Each run | 0.201 s | 28,606,618 | 0.52 GiB |
| Candidate-specific LODO | Each run | 1.928 s | 52,118,600 | 7.76 GiB |
| **LODO/shared scoring ratio** | Each run | **9.59x** | 1.82x | **14.9x** |

For a single one-off execution, baseline build plus shared scoring is 2.967 s, so we do not claim that the shared design is always faster from an empty state. Its advantage is amortized reuse: from the second scoring run onward it is faster overall; repeated detectors, sensitivity tests, ranking, and evidence exports reuse the same baseline. These measurements support retaining our shared-baseline protocol while reporting LODO as a robustness check.

## Supplementary Top-k Ranking Evaluation

The real-world Top-10 event labels are not used to compare incomparable raw score scales. All four methods score the same 836,153 quality-controlled area-user observations. Top-k evaluation is restricted to the same 310,028 positive deviations with user count above the context median; no IQR threshold defines this common ranking set. Raw method scores are not directly compared because they have different scales. For method M, consensus C_k is the mean within-method percentile assigned by the other three rankings to the items in Top-k(M). Effect size is the median ratio of observed user count to context median.

| Method | Other-method percentile (%) k=10 / 20 / 100 | Median fold change k=10 / 20 / 100 |
|---|---|---|
| Proposed IQR | 98.94 / 99.25 / 99.35 | 36.45 / 19.87 / 12.58 |
| Z-score | 99.68 / 99.52 / 99.47 | 13.54 / 10.86 / 10.29 |
| Median/MAD | 98.92 / 98.25 / 96.28 | 26.06 / 19.80 / 12.83 |
| Isolation Forest | 44.98 / 44.86 / 46.61 | 1.06 / 1.06 / 1.06 |

| Pairwise top-k overlap | k=10 | k=20 | k=100 |
|---|---|---|---|
| IQR vs. MAD | 70% | 80% | 80% |
| IQR vs. z-score | 0% | 0% | 74% |
| MAD vs. z-score | 10% | 5% | 65% |
| Isolation Forest vs. any statistical method | 0% | 0% | 0% |

The top-10 and top-20 orderings are method-sensitive, whereas the three statistical methods converge substantially by k=100. Isolation Forest optimizes a different multivariate notion of anomaly and prioritizes high-volume contexts whose relative user increase is only about 6%. The proposed IQR ranking combines strong relative effect size with high top-100 agreement. This table supports ranking suitability; a comparative accuracy claim across methods would require one independently labeled union of candidates returned by all methods.

## Supplementary GEO-System and Dataset Description

For completeness, we consolidate our operational source description and the measured dataset statistics needed to answer the reviewers below.

| Item | Verified description / measured value |
|---|---|
| **Record origin** | LTE eNodeB CellTrace: Layer-3 messages and measurement events exchanged between the UE and radio network, including serving/neighbor cells, radio measurements, handovers, and session setup/release context. |
| **Vendor extraction** | Nokia: XML or DAT; Ericsson: GBP or BIN; Huawei iMaster MAE: TRC or SIG. In the operator configuration, each eNodeB trace file is exported every 15 minutes. |
| **Standards context** | The Subscriber and Equipment Trace framework is covered by 3GPP TS 32.421 (trace concepts and requirements), TS 32.422 (trace control and configuration management), and TS 32.423 (trace data definition and management). MDT radio-measurement collection is specified in 3GPP TS 37.320 (overall description; Stage 2). TS 23.205 concerns a bearer-independent circuit-switched core network and is therefore not cited as a CellTrace/MDT basis. |
| **Positioning** | Use UE-provided GPS when available; otherwise estimate location from path loss to at least three nearby eNodeBs. MDT may increase the availability of positioning measurements. |
| **Cadence versus sampling** | Fifteen minutes is file-delivery cadence, not a fixed UE sampling interval. Records remain event-driven. The measured user-block record-count distribution has P50/P90 = 30/266 observations. |
| **Coverage** | The three principal collection regions are Hanoi, Da Nang, and Ho Chi Minh City. Our audit also found subscriber observations in Hue and northern peri-urban locations, so strict municipal boundaries require confirmation from a complete source-cell inventory. Time span: 23 Jan 2026 15:15 to 6 Feb 2026 15:14; 18,722,543,634 usable records; 9,359,862 anonymized UEs; 58,148 active H3 resolution-8 areas. |
| **Implementation** | ClickHouse 26.1.3.52 on the evaluation server with 75 vCPUs and 230 GiB RAM. |
| **Location-quality metadata** | The GEO record contains a field q, reported in metres, for the system-estimated location uncertainty. Lower q denotes lower estimated uncertainty. The measured sample distribution is reported immediately below. This metadata is not an independently GPS-validated true-error measurement. |
| **Movement QC** | Correct H3(latitude, longitude) ordering; at least 5 points and 5 H3 cells; at least 8 reference days; reference IQR <= 10 cells; all user-analysis points retained; LODO sensitivity reported separately. |
| **Area processing** | Area signals are aggregated in coarser H3 resolution-8 cells. All imported blocks are used without a full-block completeness filter. |
| **Privacy and release** | UE identifiers are anonymized. Subscriber-level GEO data are operationally confidential and cannot be publicly released. Geospatial evidence is limited to authorized analysts with access, retention, and purpose controls. |

### Measured Distribution of GEO Location-Quality Field q

The following counts were measured directly for the study interval. q is recorded in metres by the GEO system; smaller values indicate lower estimated location uncertainty.

| Dataset stage | Total samples | q interval (m) | Samples in interval (%) |
|---|---|---|---|
| Raw usable GEO input | 18,722,543,634 | q <= 100 | 3,463,079,623 (18.50%) |
| Raw usable GEO input | 18,722,543,634 | 100 < q <= 200 | 6,901,872,588 (36.86%) |
| Raw usable GEO input | 18,722,543,634 | 200 < q <= 500 | 5,738,885,758 (30.65%) |
| Raw usable GEO input | 18,722,543,634 | 500 < q <= 1,000 | 1,932,604,296 (10.32%) |
| Raw usable GEO input | 18,722,543,634 | q > 1,000 | 686,101,369 (3.66%) |
| User-analysis input | 2,454,297,496 | q <= 100 | 416,085,603 (16.95%) |
| User-analysis input | 2,454,297,496 | 100 < q <= 200 | 837,158,945 (34.11%) |
| User-analysis input | 2,454,297,496 | 200 < q <= 500 | 792,057,169 (32.27%) |
| User-analysis input | 2,454,297,496 | 500 < q <= 1,000 | 316,528,597 (12.90%) |
| User-analysis input | 2,454,297,496 | q > 1,000 | 92,467,182 (3.77%) |

Measured q quantiles (P50/P75/P90/P95/P99) are 181/325/615/867/1,841 m for the raw usable GEO input and 196/368/668/891/1,766 m for the user-analysis input. These values characterize the system-reported location-quality field and should not be interpreted as an independently GPS-validated true-error distribution.